16.0 Summative Assessment: Network Systems Characteristics

- Reflect on what you learned in each lab and any challenges you faced.
  1. I selected Lab 28 because it brings together several skills required to build and maintain a network system. I had to analyze network requirements, calculate an efficient VLSM addressing scheme, configure addressing on routers, switches, and hosts, and verify connectivity across the completed network.
  2. This lab strengthened my understanding of how subnetting moves from a calculation exercise into actual network implementation. The most important part for me was connecting the VLSM design to the device configurations. I also gained more practice recognizing that subnet size must be based on the number of usable host addresses, not simply the total number of addresses in the subnet.
  - YouTube Video Walkthrough:
    - **Lab 28 - Design and Implement VLSM Addressing (YouTube)**
  - Lab:
    - **Lab 28| Cisco Networking Labs - Design and Implement a VLSM Addressing Scheme**

Packet Tracer - Design and Implement a VLSM Addressing Scheme

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Police	**G0/0**	**10.1.1.97**	**255.255.255.240 /28**	N/A
	G0/1	**10.1.1.65**	**255.255.255.224 /27**	N/A
	S0/0/0	**10.1.1.121**	**255.255.255.252 /30**	N/A
Schools	**G0/0**	**10.1.1.113**	**255.255.255.248 /29**	N/A
	G0/1	**10.1.1.1**	**255.255.255.192 /26**	N/A
	S0/0/0	**10.1.1.122**	**255.255.255.252 /30**	N/A
PD-1	**VLAN 1**	**10.1.1.98**	**255.255.255.240 /28**	**10.1.1.97**
PD-2	**VLAN 1**	**10.1.1.66**	**255.255.255.224 /27**	**10.1.1.65**
PS-101	**VLAN 1**	**10.1.1.114**	**255.255.255.248 /29**	**10.1.1.113**
PS-115	**VLAN 1**	**10.1.1.2**	**255.255.255.192 /26**	**10.1.1.1**
PD-1-11	**NIC**	**10.1.1.110**	**255.255.255.240 /28**	**10.1.1.97**
PD-2-23	**NIC**	**10.1.1.94**	**255.255.255.224 /27**	**10.1.1.65**
PS-101-87	**NIC**	**10.1.1.118**	**255.255.255.248 /29**	**10.1.1.113**
PS-115-12	**NIC**	**10.1.1.62**	**255.255.255.192 /26**	**10.1.1.1**

Objectives

In this lab you will design a VLSM addressing scheme given a network address and host requirements. You will configure addressing on routers, switches, and network hosts.

- Design a VLSM IP addressing scheme given requirements.
- Configure addressing on network devices and hosts.
- Verify IP connectivity.
- Troubleshoot connectivity issues as required.

Background / Scenario

You have been asked to design, implement, and test an addressing scheme for a customer. The customer has given you the network address that is suitable for the network, the topology, and the host requirements. You will implement and test your design.

Instructions

You have been given the network address **10.1.1.0/24** by your customer. The host address requirements are:

Note: Some of the information is provided in the Packet Tracer instructions for your topology.

Requirements

Host Requirements:

LAN	Number of Addresses Required
PD-1 LAN	11
PD-2 LAN	28
PS-101 LAN	5
PS-115 LAN	47
WAN	2

Design Requirements

- Create the addressing design. Follow guidelines provided in the curriculum regarding the order of the subnets.
- The subnets should be contiguous. There should be no unused address space between subnets.
- Provide the most efficient subnet possible for the point-to-point link between the routers.
- Document your design in a table such as the one below.

Subnet Description	Number of Hosts Needed	Network Address/CIDR	First Usable Host Address	Broadcast Address
PS-115 LAN **[/26]**	47	**10.1.1.0**	**10.1.1.1**	**10.1.1.63**
PD-2 LAN **[/27]**	28	**10.1.1.64**	**10.1.1.65**	**10.1.1.95**
PD-1 LAN **[/28]**	11	**10.1.1.96**	**10.1.1.97**	**10.1.1.111**
PS-101 LAN **[/29]**	5	**10.1.1.112**	**10.1.1.113**	**10.1.1.119**
WAN **[/30]**	2	**10.1.1.120**	**10.1.1.121**	**10.1.1.123**

Note: You will configure addressing on **all** devices and hosts in the network.

- Assign the first usable IP addresses in the appropriate subnets to Police for the two LAN links and the WAN link.
  - **G0/0 10.1.1.97,**
  - **G0/1 10.1.1.65,**
  - **S0/0/0 10.1.1.121**

- Assign the first usable IP addresses in the appropriate subnets to Schools for the two LANs links.
    - **G0/0 10.1.1.113,**
    - **G0/1 10.1.1.1**
- Assign the last usable IP address for the WAN link.
    - **S0/0/0 10.1.1.122**
- Assign the second usable IP addresses in the appropriate subnets to the switches.
    - **PD-1: 10.1.1.98,**
    - **PD-2 10.1.1.66,**
    - **PS-101 10.1.1.114,**
    - **PS-115 10.1.1.2**
- The switch management interface should be reachable from hosts on all of the LANs.
- Assign the last usable IP addresses in the appropriate subnets to the hosts.
    - **PD-1-11: 10.1.1.110,**
    - **PD-2-23 10.1.1.94,**
    - **PS-101-87 10.1.1.118,**
    - **PS-115-12 10.1.1.62**

If the addressing design and implementation are correct, all hosts and devices should be reachable over the network.